

2(a)	S = protein ; A amino acid R protein coat I capsomere T = capsid ; A protein coat R capsomere	2
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Question	Answer	Marks
2(b)	170 (nm) ;	1
2(c)	replication / synthesis / AW, of (viral) DNA ; I replication of genes A <i>stated function of DNA polymerase</i> I unwinding of DNA e.g. forms phosphodiester bonds joins / AW, (adjacent DNA) nucleotides R joins complementary nucleotides I joins sugar phosphate backbone	1
2(d)	presence of (viral), envelope / (outer) membrane / phospholipid bilayer ; ref. fluid / flexible ; ref. to preparation (for electron microscopy) ; e.g. distortion / squashed / artefact I ref. to use of the microscope e.g. electron beam affects AVP ; e.g. <i>suggestion of variation in <u>tegument</u> protein, number / arrangement / organisation / AW</i>	2

Question	Answer	Marks
2(e)	<i>similarity, one mark:</i> both have <u>DNA</u> (double stranded, naked / no histones) ; A <u>DNA</u> plus neutral feature e.g. not surrounded by a nuclear envelope I both have genes I components of DNA e.g. deoxyribose / nucleotides / phosphates R both have, single stranded / circular, DNA R both have, DNA free in the cytoplasm / DNA and RNA <i>difference, one mark:</i> bacterial DNA, circular / closed loop (v HCMV linear) I ref. to naked DNA R if comparison for HCMV is incorrect e.g. multilayered or bacteria (can also) have, plasmid DNA / plasmids R if implied that bacterial DNA is <u>only</u> plasmid DNA or bacterial DNA (also) has genes for, metabolism / AW ora or bacterial DNA free in <u>cytoplasm</u> but HCMV DNA surrounded by, capsid / protein coat / T ;	2

Question	Answer	Marks
2(f)	I transcription and translation stops any three from: 1 no, S phase / G₂ phase / mitosis / cytokinesis ; A remains in (G₁) interphase A synthesis phase <i>for</i> S phase I named mitotic stages <i>look for CON</i> e.g. 'no S-phase, so mitosis only has one chromatid' 2 (semi-conservative) DNA, replication / synthesis, does not occur ; A sister / identical, chromatids, do not form R if a phase is stated and it is not given as the S-phase 3 failure to pass checkpoints / does not reach checkpoints ; A ref. to no error checking occurs (<i>in context of checkpoints</i>) I proofreading / error checking, (by DNA polymerase) during DNA replication 4 cell does not, grow / increase in size / increase cytoplasm ; (<i>context of preparing for mitosis / cytokinesis</i>) A grows to normal (non-dividing) size 5 <i>idea that</i> the cell does not, duplicate / synthesise, organelles / extra materials (<i>context is for, cytokinesis / new cells</i>) ; A centrioles do not replicate A microtubules not synthesised R if stated as occurring in S phase (even if e.g. G₂ given as well)	3

Question	Answer	Marks
2(g)	<i>allow virus or pathogen for HCMV</i> <i>look for ora in context of healthy immune system</i> any four from: 1 ref. to time delay in immune response (compared to normal) ; e.g. takes a longer time for, immune system to respond slower (secondary) immune response immune response similar to primary response 2 more time / easier / AW, for HCMV to infect cells / replicate / spread (so cause damage) ; AW R ref. to HCMV as, cells that reproduce / viruses that carry out mitosis <i>impact following primary response and after viral activation</i> 3 few(er), immune system cells / named, (in circulation) ; I fewer white blood cells e.g. fewer, macrophages / neutrophils A phagocytes fewer (B- and T-) memory cells A no memory cells (<i>no longer present from primary response</i>) fewer, (B/T,-) lymphocytes / T-helper cells / T-killer (or cytotoxic) cells <i>context for lymphocytes could be reduced number circulating at time of viral activation or owing to poor secondary response(compared to normal)</i> fewer / no, plasma cells 4,5 consequence of fewer immune system cells ;; <i>two marks</i> e.g. less chance of / AW, encounter with / recognition of / activation by, (non-self / foreign) antigen / HCMV / virus / pathogen / infected cell (fewer memory cells) ref. reduced clonal expansion / AW (so fewer lymphocytes / plasma cells)	4

Question	Answer	Marks
2(g)	(fewer plasma cells) lower concentration of antibodies A no antibodies (as no plasma cells) A longer time for antibody, production / concentration to build up (fewer T-helper cells) less cytokine released A effect of cytokine <i>if spelling of cytokine</i> rejected (fewer T-killer cells) fewer infected cells killed / A perforin / granzymes / hydrogen peroxide, not released (fewer phagocytes) less phagocytosis for antigen presentation	

Question	Answer	Marks
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